

## Examination ticket № 92

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### 1) Lyapunov stability. The second Lyapunov theorem.

**Answer:**

Let  $p = 0$  is an equilibrium point of the system and  $D \subset \mathbb{R}^n$  is an open set, including  $p =$

0. Let  $V : D \rightarrow \mathbb{R}$  is a continuously differentiable function such that

$$V(0) = 0 \text{ and } V(x) > 0 \text{ in } D \setminus \{0\}$$

If  $\dot{V}(x) \leq 0$  in  $D$  due to the equation of the system, then  $p = 0$  is stable.

If  $\dot{V}(x) < 0$  in  $D \setminus \{0\}$  by virtue of the equation of the system, then  $p = 0$  is asymptotically stable.

The function  $V(x)$  is called Lyapunov function.

### 2) Consider the system

$$\dot{x}_1 = -3x_1 + 7x_1^2$$

$$\dot{x}_2 = -x_2 + 2x_3^2$$

$$\dot{x}_3 = x_3 - 5x_1^2$$

Linearized system near zero equilibrium.

**Answer:**

Step 1: Find the equilibrium points. The zero equilibrium is clearly at  $(0,0,0)$  because when set the right-hand sides to zero:

$$\text{First: } -3x_1 + 7x_1^2 = x_1(-3+7x_1)=0 \Rightarrow x_1=0 \text{ or } x_1=3/7.$$

$$\text{Second: } -x_2 + 2x_3^2=0 \Rightarrow x_2 = 2x_3^2.$$

$$\text{Third: } x_3 - 5x_1^2=0 \Rightarrow x_3 = 5x_1^2.$$

So if  $x_1=0$ , then  $x_3=0$ , and then  $x_2=0$ . So  $(0,0,0)$  is an equilibrium.

Step 2: Linearize around  $(0,0,0)$ . The linearization is given by the Jacobian matrix evaluated at the equilibrium.

Let

$$\dot{x}_1 = f_1(x_1, x_2, x_3) = -3x_1 + 7x_1^2,$$

$$\dot{x}_2 = f_2(x_1, x_2, x_3) = -x_2 + 2x_3^2,$$

$$\dot{x}_3 = f_3(x_1, x_2, x_3) = x_3 - 5x_1^2.$$

Compute partial derivatives:

For f1:

$$\frac{\partial f_1}{\partial x_1} = -3 + 14x_1, \text{ at } (0,0,0) : -3,$$
$$\frac{\partial f_1}{\partial x_2} = 0, \frac{\partial f_1}{\partial x_3} = 0.$$

For f2:

$$\frac{\partial f_2}{\partial x_1} = 0, \frac{\partial f_2}{\partial x_2} = -1,$$
$$\frac{\partial f_2}{\partial x_3} = 4x_3, \text{ at } (0,0,0) : 0.$$

For f3:

$$\frac{\partial f_3}{\partial x_1} = -10x_1, \text{ at } (0,0,0) : 0,$$
$$\frac{\partial f_3}{\partial x_2} = 0, \frac{\partial f_3}{\partial x_3} = 1.$$

So the Jacobian matrix at (0,0,0) is:

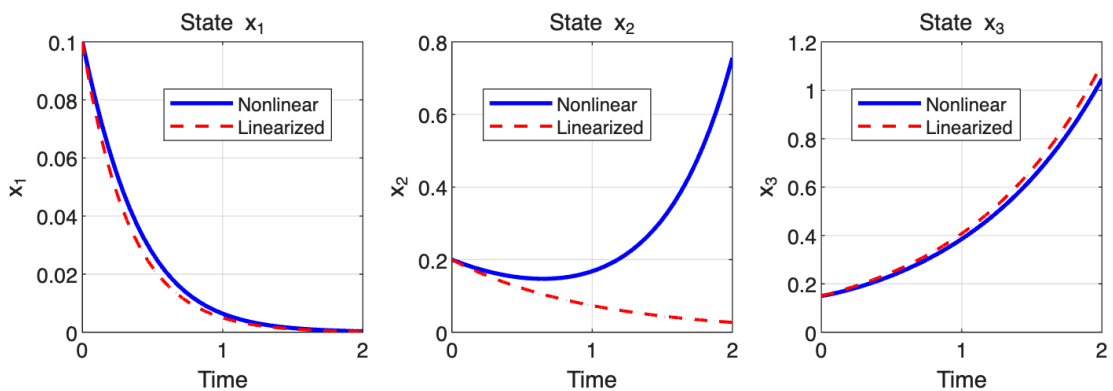
$$A = \begin{pmatrix} -3 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Thus, the linearized system is:

$$\dot{x}_1 = -3x_1$$
$$\dot{x}_2 = -x_2$$
$$\dot{x}_3 = x_3$$

This can also be written in matrix form as  $\dot{\mathbf{x}} = A\mathbf{x}$ .

Comparison: Nonlinear vs Linearized System



Phase Space Trajectory

